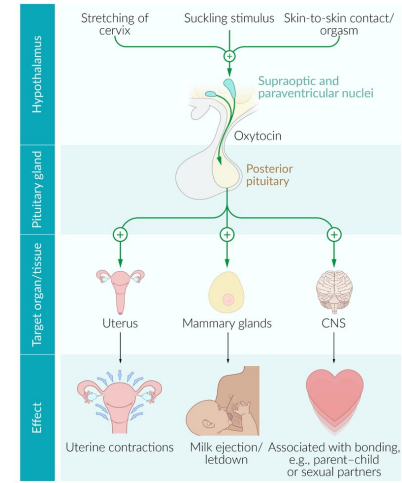
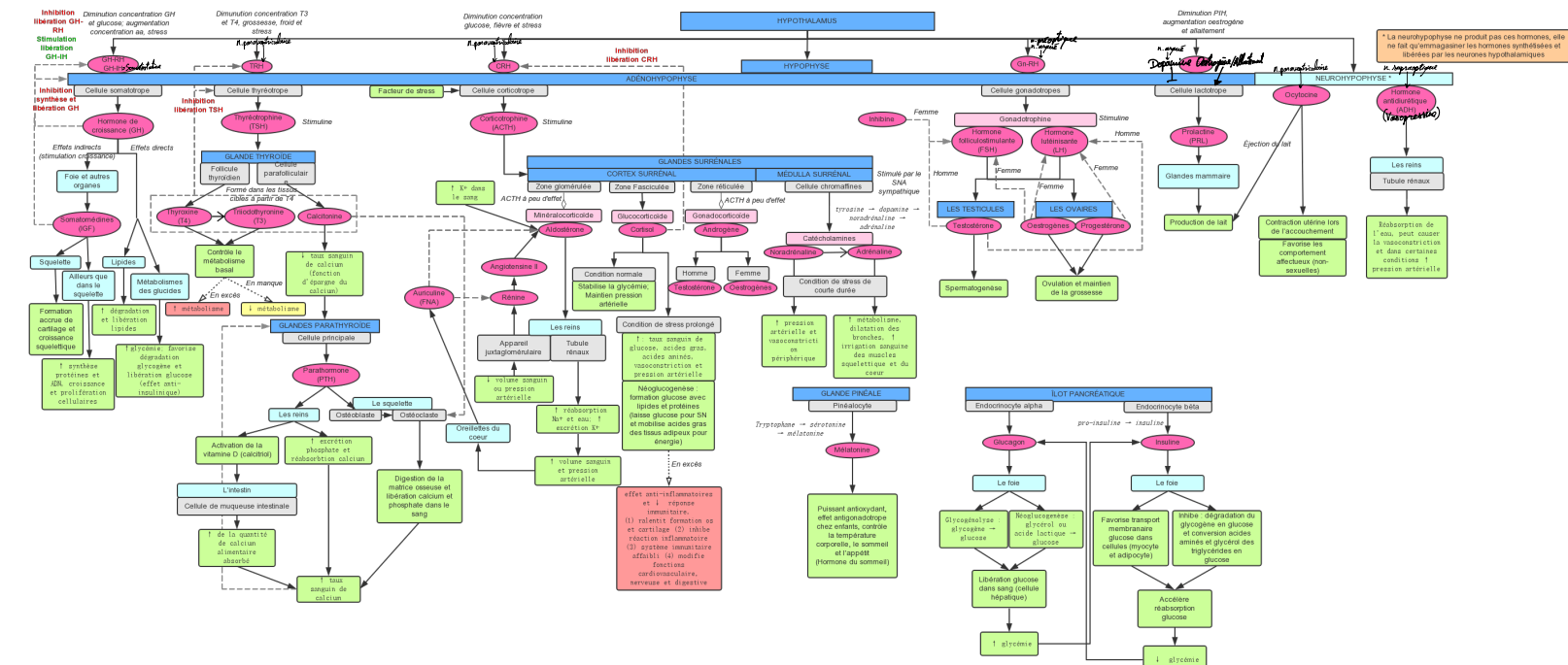
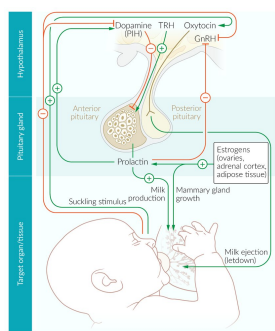
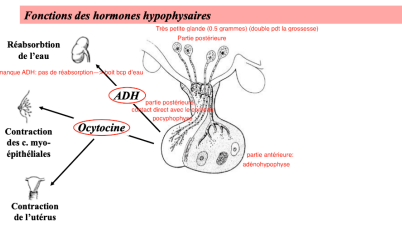
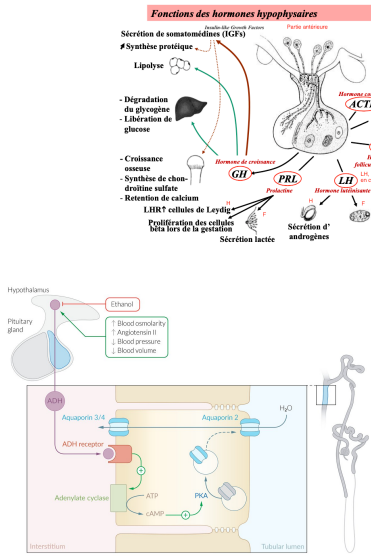
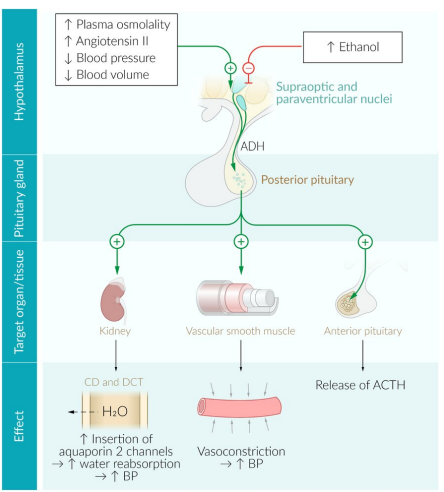


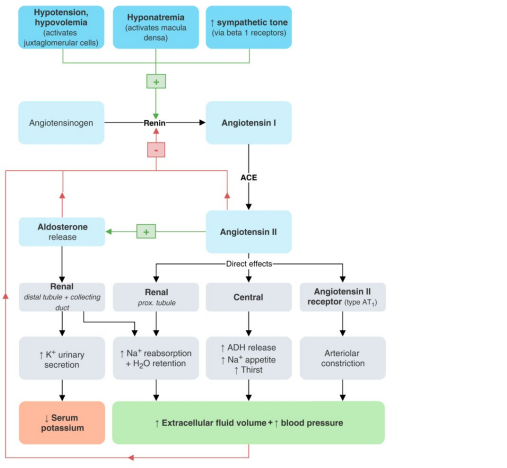
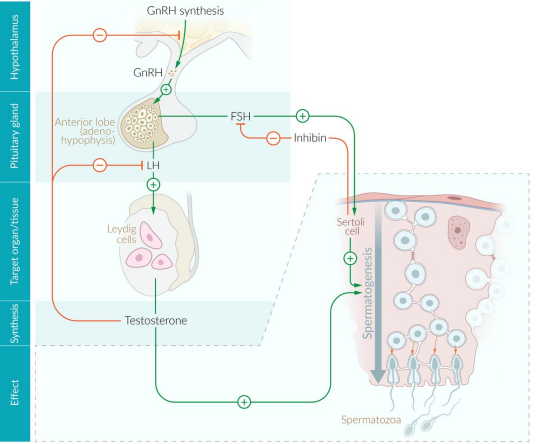
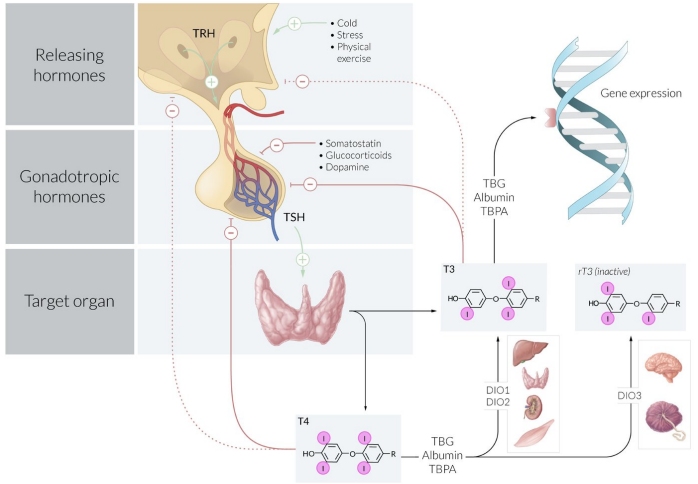
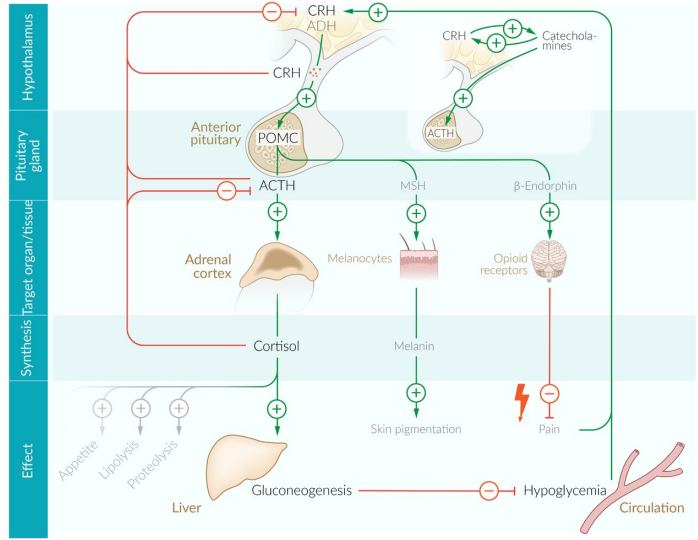
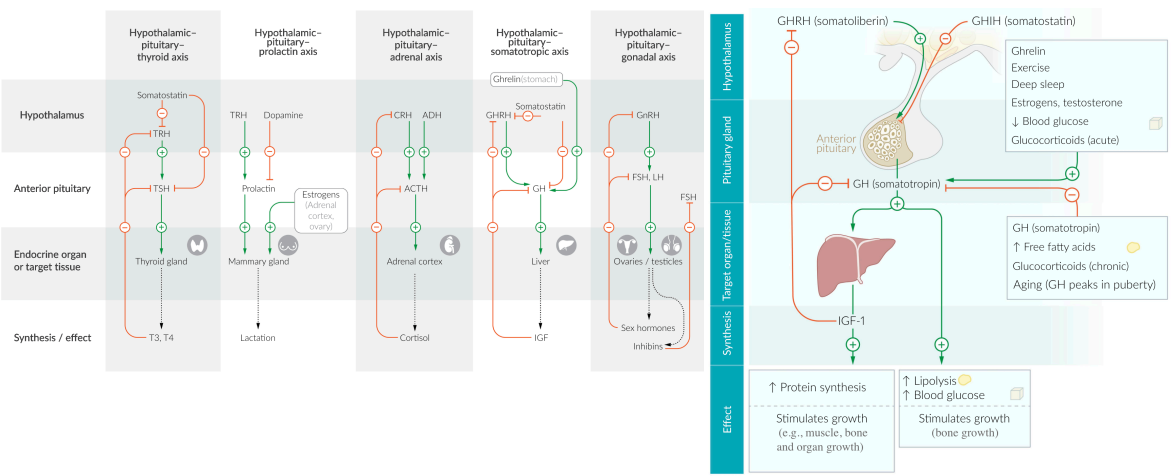


*Oxytocine
ADH
Prolactine
Glucagon
Insuline*

*Testostérone
Progestérone
Œstrogène
Glucocorticoïde
Minéralocorticoïde*

	Peptides & protéines	H stéroïdiennes	Catécholamines	H thyroïdiennes
Synthèse	Synthèse protéique	Modification cholestérol	Modification tyrosine	Modification tyrosine
Solubilité	Hydrophiles	Lipophiles	Hydrophiles	Lipophiles
Stockage	Granules de sécrétion	Absent	Granules de sécrétion	Colloïde
Sécrétion	Exocytose contrôlée	Diffusion	Exocytose contrôlée	Diffusion
Transport	Libres / liés aux protéines	Liés aux protéines	Libres	Liés aux protéines
Demi-vie	Courte - moyenne (4 - 40 - 170 min)	Moyenne	Très courte (< 3 minutes)	Longue (20 h - 7 j)
Récepteurs	Membrane	Cytosol	Membrane	Noyau
Vitesse de réponse	Rapide	Lente	Très rapide	Lente
Durée de l'effet	Courte	Longue	Très courte	Longue





Organe	fonction	action
Hypothalamus	Hypothalamus	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Hypophyse	Hypophyse	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Thyroïde	Thyroïde	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Parathyroïde	Parathyroïde	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Surrénales	Surrénales	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Pancréas	Pancréas	Contrôle la sécrétion de l'axe hypothalamo-pituitaire
Gonades	Gonades	Contrôle la sécrétion de l'axe hypothalamo-pituitaire

Organe	fonction	action	dosage clinique
Hypothalamus	Hypothalamus	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Hypophyse	Hypophyse	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Thyroïde	Thyroïde	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Parathyroïde	Parathyroïde	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Surrénales	Surrénales	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Pancréas	Pancréas	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance
Gonades	Gonades	Contrôle la sécrétion de l'axe hypothalamo-pituitaire	Dosage en cas de déficit ou excès d'hormone de croissance

[illegible]

Symptômes principaux

- apathie, depression, léthargie
- **intolérance au froid**
- anorexie
- **constipation**
- faiblesse musculaire, douleurs musculaires
- **perte de cheveux**
- **ongles cassants**
- syndrome du carpien
- oligo-aminorrhéique ou ménorragie, fertilité ↓

HYPOTHYROIDISME

- Intolérance to Cold
- Receding Hairline
- Facial & Eyelid Edema
- Gull-Blank Expression
- Dry Skin
- Thick Tongue
- Hoarse Voice
- Anemia
- Brittle Nails & Fingers
- Constipation
- Menstrual Disturbances
- Dry Discharge
- Subnormal Temp
- Fatigue

- nervosité, irritabilité
- transpirations, intolérance à la chaleur
- palpitations
- appétit augmenté
- selles plus fréquentes
- faiblesse musculaire, fatigabilité
- perte de cheveux
- ongles cassants
- ♀: oligoménorrhée, aménorrhée, infertilité

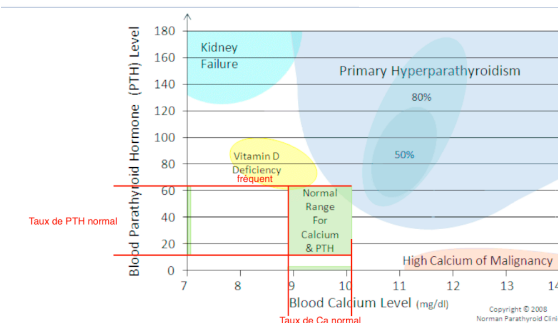
• prise de poids
 • bradycardie
 • hypertension diastolique
 • **reflexes hypovifs** (*signe de Babinski normalisé*)
 • peau sèche, épaisse
 • oedèmes (e.g., oedème, facial)
 • cheveux crispés
 • voix rauque
 • macroglossie, oedème des muqueuses

Hair Loss
 Apathy
 Lethargy
 Dry Skin
 (Cracks & Peeling)
 Muscle Aches
 & Weakness
 Constipation
 Receding Hairline
 Facial & Scleral Edema
 Cullen's Sign Expression
 Extreme Fatigue
 Puffy Tongue
 Slow Speech
 Anorexia
 Brittle Nails
 & Hair
 Menstrual Disturbances
 Luteal Manifestations
 Subnormal Temp
 Bradycardia
 Nighttime Gain
 2-3°C

- perte pondérale (analyse 1° aspect) (catabolisme)
- tachycardie, arythmie
- hypertension systolique
- réflexes hypervifs → sphère excitée (cf. 1° aspect)
- peau humide, hyperémique
- cheveux fins, droits
- signes oculaires (Cf. ci-après)

- **Paresthésies (fourmillements)**
 - doigts, oreilles, lèvres/péri-puéal
- **Douleurs ou crampes musculaires**
 - jambes, pieds, abdomen, visage
- **Contractions ou spasmes musculaires**
 - région péribuccale, mains, bras, larynx
- **Fatigue, faiblesse**
- **Perte de cheveux**
- **Peau sèche**
- **Ongles cassants**
- **Dépression ou anxiété**

- **Néphrolithiase**
- Polyakiurie
- **Nausées, vomissements, manque d'appétit**
- Douleurs abdominales
- **Fatigabilité, faiblesse musculaire**
- **Dépression, troubles de la mémoire**
- Douleurs osseuses et articulaires
- Plaintes non-spécifiques sans cause évidente
- **Ostéoporose** (radiologique ou fracturaire)



(angiotensine) **CLINIQUE INSUFFISANCE**
SURRENALIENNE - *Maladie d'Addison*

- **Asthénie croissante**, majeure (cortisol)
- **Maigrissement et anorexie**
- **Hypotension et hypoglycémie**
- **Troubles digestifs** :
 - douleurs abdominales, diarrhées
 - peu spécifiques annonciateurs d'une crise aiguë
- **Mélanodermie** = hyperpigmentation au niveau des zones découvertes ou de frottement : la **main addisonienne**
 - ⇒ Signe l'origine périphérique (**ACTH élevée**)

INSUFFISANCE SURRENALE AIGUE :
!! URGENCE VITALE !!

- Association d'un **collapsus cardiovasculaire**, d'une **adynamie** extrême et de **nausées et vomissements profus**

- Au moment de la découverte ou en cas de **stress majeur** (trauma, chirurgie, maladie) ou **oubli du traitement substitutif** (*ne pas oublier*)

2- Syndrome de Cushing

Hyper production d'hormones glucocorticoïdes (**cortisol**) et d'androgènes par les glandes surrénales *Cause psychologique ou iatrogène, ACTH*

≠ • Différent de la **Maladie de Cushing** où la **cause est hypophysaire** ... adénome sécrétant de l'ACTH

Manifestations cliniques du syndrome de Cushing

Les plus spécifiques		Les plus aspécifiques
• Hémorragies spontanées	• Obésité centrale avec faiblesse des ceintures	• DM
• Myopathie proximale	• HTA	• Dépression
• Strabisme accommodatif	• Obésité	• Insomnie
• Violences	• <i>Non possible</i>	• Psychose
	• Oligo-aminosémie	• Hirsutisme
	• Faciès lunaire	• Infections fongiques
	• Ostéoporose	

Diagnostique

Obésité	27
Insomnie	20
Myopathie proximale	17
Violence	16
HTA	15

Donnée from: Rhee, Si, Ueda, D. Cauter's clinical endocrinology: diagnostic evaluation and management of endocrine disorders, 10th edition, 2016

Sy. de Cushing
CLINIQUE: COMPLICATIONS
(Cortisol = $\uparrow$ hyperglycémie)

- HTA et Diabète
- Dyslipidémie
- Infections
- Ostéoporose
- Hirsutisme, oligoménorrhée
- Troubles psychiques : 50 % des cas, syndrome dépressif ou délire...

Carcinoid syndrome

The diagram shows two female figures representing different stages of breast cancer. The left figure represents a primary tumor, and the right figure represents metastasis. A legend at the top indicates that purple shading represents the 'Primary tumor' and red shading represents 'Metastasis'.

Primary Tumor (Left Figure):

- Hormonal symptoms:** Indicated by a purple dot on the upper chest area.
- First-pass metabolism:** Indicated by a red 'X' in the liver area.
- No hormonal symptoms, abdominal pain or intestinal obstruction may occur:** Indicated by a purple dot in the abdominal area.

Metastasis (Right Figure):

- Dyspnea:** Indicated by a red dot on the upper chest area.
- Diaphoresis (with flushing):** Indicated by red dots on the face.
- Cutaneous flushing:** Indicated by red shading on the face.
- Hypotension and tachycardia (with flushing):** Indicated by red shading on the chest area.
- Coronoid heart disease:** An inset image shows a heart with a red area indicating disease.
- Abdominal cramps:** Indicated by a red dot in the abdominal area.
- Diarrhea:** Indicated by a red dot in the abdominal area.

Diagram illustrating various complications of diabetes mellitus, labeled on a human figure:

- Retinopathy
- Cerebrovascular disease
- CHD
- Monckeberg arteriosclerosis (medial calcific sclerosis)
- Gastroparesis
- Nephropathy
- Glomerulosclerosis (diffuse or with Kimmelstiel-Wilson nodules)
- Erectile dysfunction, bladder dysfunction
- PAD: gangrene and ulceration
- Peripheral neuropathy
- Diabetic foot

A diagram of a human arm and hand, illustrating the progression of atherosclerosis. Labels with lines pointing to different parts of the arm indicate the following conditions:

- Cerebrovascular disease**: Points to the head/neck area.
- CHD** (Coronary Heart Disease): Points to the chest area.
- Mönckeberg arteriosclerosis (medial calcific sclerosis)**: Points to a red, inflamed area on the forearm.
- PAD: gangrene and ulceration** (Peripheral Artery Disease): Points to the hand and fingers.

Criteria

Clinical features

• glucose (patients with diabetes: < 70 mg/dL; patients without diabetes: < 50 mg/dL)

• improve with normalization of blood glucose

Causes

• hypoglycemia: acute illness (e.g., infection), related (e.g., accidental overdose of insulin), drug, fasting, exercise

• hypoglycemia: chronic illness (e.g., sepsis), iatrogenic, accidental or intentional insulin overdose, hyperinsulinism (e.g., insulinoma)

Diagnosis

• low glucose levels

• common causes of hypoglycemia (e.g., infections)

• common causes were identified, consider testing (e.g., fasting test)

Management

• administration of oral glucose or IV dextrose

Neurogenic/autonomic symptoms

- Anxiety, pallor, sweating
- Tachycardia, palpitations
- Hunger, nausea, vomiting



Neuroglycopenic symptoms

- Impaired consciousness
- Agitation
- Fatigue
- Seizures

—se

Hyponatremia

Definition
Low serum sodium concentration < 135 mEq/L (and in hypotonic/true hyponatremia also low serum osmolality < 280 mOsm/kg H₂O).
Onset: acute (< 48 hours) or chronic (> 48 hours or duration unknown)

Etiology
Hypovolemic: e.g., GI/renal/dermal losses, third-spacing, hemorrhage
Euvolemic: e.g., SIADH, low salt intake, water intoxication
Hypervolemic: e.g., renal failure with oliguria, congestive heart failure

Diagnostics
Serum sodium (corrected for hyperglycemia) and osmolality
Further serum/urine studies to identify etiology

Treatment
Acute or severe symptoms: Correct sodium quickly with hypertonic saline, then treat underlying cause.
Chronic without severe symptoms: Correct sodium slowly (cause-specific treatment).

Complications
Secondary to hyponatremia: e.g., cerebral edema
Treatment-associated: e.g., osmotic demyelination syndrome

Severe symptoms (usually acute hyponatremia)

Stupor or coma
Seizures
Ataxia
Respiratory failure
(severity correlates with degree of cerebral edema)

Mild/moderate symptoms

Headache, dizziness, forgetfulness, malaise
Nausea, vomiting

Signs of volume status

Hypovolemia: e.g., dry mucous membranes, orthostatic hypotension, oliguria
Hypervolemia: e.g., ascites, effusions, peripheral edema

Hypermnatremia

Definition
Elevated serum sodium concentration (> 145 mEq/L)

Epidemiology
~1% of hospitalized patients ≥ 60 years of age

Etiology
Inadequate fluid intake
Fluid loss
Excessive sodium intake
Sodium retention

Diagnosis
Clinical assessment of volume status
Serum and urine sodium concentration
Serum and urine osmolality

Treatment
Correct underlying condition
Calculate and correct free water deficit

Complications
Secondary to hypernatremia: e.g., osmotic demyelination, rhabdomyolysis
Secondary to IV fluid correction: e.g., cerebral edema, hypervolemia

Mild

Thirst, dry mucous membranes
Dry skin

Moderate

Confusion, irritability, lethargy
Muscle weakness
Hyperreflexia

Severe (Na > 160)

Altered mental status
Seizure, stupor, coma

Chronic (> 48 hours)

Commonly asymptomatic or mild symptoms

Rickets

Etiology

- Vitamin D deficiency due to:
 - Insufficient synthesis due to low UV radiation exposure (e.g., northern climates) and/or dark skin
 - Insufficient oral intake (e.g., in infants who are exclusively breastfed)
 - Malabsorption
 - Defective vitamin D metabolism

Pathophysiology
Vitamin D deficiency → defective mineralization of osteoid and growth plates

Diagnostics
↓ Vitamin D, ↓ serum Ca²⁺, ↑ PTH, ↓ PO₄³⁻, ↑ ALP

Treatment

- Dietary supplementation (vitamin D, calcium)
- Treatment of underlying cause

Bone abnormalities

Delayed fontanelle closure
Craniotabes
Frontal bossing
Rachitic rosary
Enlarged epiphyses
Fragile bones
Normal bone
Bowing legs

© AMBOSS GmbH

Hypokalemia

Definition
Hypokalemia: serum potassium (K⁺) < 3.5 mEq/L
Severe hypokalemia: serum K⁺ < 2.5 mEq/L

Etiology
Gastrointestinal loss: e.g., due to vomiting, diarrhea
Renal loss: e.g., due to renal diseases, medications
Intracellular shift: e.g., due to alkalosis, insulin excess
Insufficient intake: e.g., due to eating disorders

Diagnostics
Electrolytes and kidney function, blood gas, urinary K⁺
ECG (e.g., T-wave flattening, ST depression, U waves, dysrhythmias)

Treatment
Potassium chloride (KCl) repletion
Management of concurrent electrolyte abnormalities (KCl repletion is ineffective in untreated hypomagnesemia)
Treatment of underlying cause

Note
Patients may be asymptomatic, especially if K⁺ > 3.0 mEq/L

Cardiovascular

Arrhythmias (palpitations, irregular pulse, syncope)
Hypotension
Constipation
Polyuria

Neuromuscular
(cramps and spasms, weakness, decreased deep tendon reflexes)

Hyperkalemia

Definition
Serum potassium (K⁺) level > 5 mEq/L

Etiology
Reduced excretion (e.g., due to renal disease or drugs)
Extracellular shift (e.g., due to insulin deficiency or cell lysis)
Increased intake (e.g., due to high-potassium diet, K⁺ containing IV fluids)
Pseudohyperkalemia (e.g., due to fist clenching during venipuncture, delayed analysis)

Clinical features
Symptoms usually occur if serum K⁺ > 7.0 mEq/L or rapidly increasing

Diagnostics
Electrolytes, glucose, kidney and liver function, complete blood count, blood gas
ECG (e.g., peaked T-waves, QRS complex widening)

Treatment
Cardiac membrane stabilization with calcium salts
Intracellular K⁺ shifting (e.g., with insulin and glucose)
Enhanced K⁺ elimination (e.g., with hemodialysis)
Treatment of underlying cause

Other features

Hyperglycemia
Symptoms of underlying cause: e.g., dehydration (in gastroenteritis), tachycardia and tremors (in alcohol withdrawal)
Fatigue

Cardiac arrhythmias, e.g.,
bradycardia, AV block, ventricular fibrillation

Nausea, vomiting

Diarrhea

Paresthesia

Muscle weakness, paralysis

↓ Deep tendon reflexes

Hypocalcemia

Definition
Mild hypocalcemia: total serum Ca²⁺ < 8.5 mg/dL or ionized Ca²⁺ < 4.65 mg/dL
Severe hypocalcemia: total Ca²⁺ ≤ 7.5 mg/dL or ionized Ca²⁺ < 3.6 mg/dL
Factitious: decreased total Ca²⁺ with normal ionized Ca²⁺ (asymptomatic)

Etiology
With low PTH: hypoparathyroidism
With high PTH: vitamin D deficiency, CKD, acute pancreatitis, hyperphosphatemia
Other: medications (e.g., loop diuretics), multiple blood transfusions, hypomagnesemia

Diagnostics
Serum total and ionized Ca²⁺, electrolytes and kidney function, ABG/VBG, serum intact PTH
ECG (e.g., prolonged QT interval)

Treatment
Calcium supplementation
Treatment of the underlying condition

Neurological

Seizure
Tetany (increased neuromuscular excitability):
Paresthesias (e.g., perioral)
Spasms (e.g., carpopedal)
Muscle cramps, stiffness, myalgia

Cardiovascular

Arrhythmias (palpitations, irregular pulse, syncope, hypotension, congestive heart failure)

Chronic manifestations

Psychiatric: e.g., anxiety, depression
Neurological: pseudotumor cerebri
Ophthalmologic: papilledema, cataracts
Dental changes: altered morphology, enamel hypoplasia

Note

Symptoms of tetany (e.g., spasms, cramps and paresthesia) are the most characteristic features of hypocalcemia.

Hypercalcemia

Definition
Mild hypercalcemia: total serum Ca > 10.5 mg/dL or ionized Ca²⁺ > 5.25 mg/dL
Severe hypercalcemia: total Ca > 14 mg/dL or ionized Ca²⁺ > 10 mg/dL

Etiology
High PTH: hyperparathyroidism (1°-3° degree), FHH
Low PTH: malignancies, granulomatous disorders
Other: medications (e.g., thiazide diuretics)

Diagnostics
Ionized Ca²⁺ or corrected serum Ca²⁺
Serum intact PTH
ECG (e.g., QT interval shortening, J waves)
Bone evaluation

Treatment
Treat underlying cause
Ensure adequate hydration, reduce calcium intake
Severe hypercalcemia: rapidly lower calcium levels with IV 0.9% NaCl (+/- diuretics) and pharmacotherapy (e.g., calcitonin, bisphosphonates, denosumab)

Neuropsychiatric

Anxiety
Depression
Fatigue
Cognitive dysfunction
Somnolence

Renal

Nephrolithiasis, nephrocalcinosis (typically Ca²⁺ oxalate stones)
Polyuria and dehydration

Osteoarticular

Myalgias
Arthralgias
Bone pain
Fractures

Note

In the setting of hypercalcemia, fractures and bone pain may be signs of excessive bone resorption (e.g., due to malignancy), whereas obtundation and coma indicate severe hypercalcemia.

Diminished muscle excitability

Cardiac arrhythmias
Muscle weakness, paresis

Gastrointestinal

Abdominal pain
Nausea and vomiting
Anorexia
Pancreatitis
Peptic ulcer disease
Constipation

Neuropsychiatric

Anxiety
Depression
Fatigue
Cognitive dysfunction
Somnolence

Renal

Nephrolithiasis, nephrocalcinosis (typically Ca²⁺ oxalate stones)
Polyuria and dehydration

Osteoarticular

Myalgias
Arthralgias
Bone pain
Fractures

Note

In the setting of hypercalcemia, fractures and bone pain may be signs of excessive bone resorption (e.g., due to malignancy), whereas obtundation and coma indicate severe hypercalcemia.

Polycystic ovary syndrome (PCOS)

Etiology
Unknown
Associated with insulin resistance and obesity

Diagnostics: Rotterdam criteria

- Oligoovulation or anovulation
- Hyperandrogenism
 - Clinical features (acne, alopecia, hirsutism) or
 - Laboratory features (↑ testosterone)
- Enlarged and/or polycystic ovaries on ultrasound examination

Treatment

- Weight loss
- No wish to conceive:
 - Combined oral contraceptives (1st line)
 - Metformin (2nd line)
- Wish to conceive:
 - Ovulation induction (1st line: letrozole)

Complications

- Cardiovascular disease
- Type 2 diabetes mellitus
- Malignancy (e.g., endometrial cancer)
- Pregnancy loss

Acromegaly

Etiology
Benign pituitary adenoma → excessive GH secretion

Pathophysiology
↑ GH → ↑ IGF-1 → overstimulation of cell growth and proliferation
Other tumor effects:
↓ Secretion of gonadotropins

Diagnostics
↑ Serum IGF-1

Treatment
First-line therapy: transsphenoidal adenomectomy

Complications
Cardiovascular disease (main cause of death), impaired glucose tolerance, carpal tunnel syndrome

Pituitary adenoma

Direct tumor effects

Headache

Bitemporal hemianopsia

Effects of excess GH/IGF-1

Coarse facial features

Wide-spaced teeth, macroglossia

Organomegaly

Large hands

Thickened skin, skin tags

Joint pain

Large feet

Paget disease of the bone (osteitis deformans)

Pathophysiology
Increased bone remodeling (↑ osteoclastic and osteoblastic activity) → replacement of lamellar bone with weak woven bone

Diagnosis

- Isolated elevation of ALP with normal calcium, phosphate, and PTH
- Urinalysis (markers of collagen degradation)
- X-ray (bone deformation with sclerotic and osteolytic lesions)
- Skeletal scintigraphy (bony lesions)

Treatment

- Indicated in active disease (↑ ALP or active foci on skeletal scintigraphy)
- First-line: bisphosphonates

Skull enlargement, leonine facies

Headaches

Hearing loss (due to ankylosis of ossicles and narrowing of the internal auditory meatus)

Bone pain

Osteosarcoma

Pathological fractures

Osteoarthritis

Bone deformation (e.g., saber shin)

Increasing hat size